UNIVERSITY OF KELANIYA / SENG 34213
FOREVER HOTEL MANAGEMENT SYSTEM

# Test Coverage Report

Six-Subsystem Coverage and Automated Test Execution

Reporting date: 9 November 2026. Evidence is identified by subsystem repository, branch, commit SHA, and GitHub Actions run date; comparisons across separate repository commits do not constitute a single deployed six-subsystem release test.

## Contents

1. Consolidated coverage overview
2. Hotel Website (HW)
3. Manager Dashboard (MAD)
4. Front Desk System (FDS)
5. Food Ordering and Service Requests (FOSS)
6. Kitchen Management System (KMS)
7. Worker Management System (WKMS)
8. Evidence interpretation and remaining actions

# 1. Consolidated coverage overview

The table records instrumented source coverage for separately tested subsystem and shared-service repositories across the forever-hotel GitHub organization as of 9 November 2026, alongside committed subsystem QA deliverable snapshots where applicable. Values are reported in percent; no unweighted cross-repository average is calculated because instrumentation tools, denominators, and excluded files differ across repositories.

| Subsystem / measured scope | Statements | Branches | Functions | Lines |
|---|---|---|---|---|
| HW - Backend Jest unit (develop 5e3cd6bf32) | 88.38 | 82.10 | 87.14 | 88.39 |
| HW - Backend Jest unit (QA snapshot 63673ba704) | 88.02 | 82.16 | 87.50 | 88.05 |
| HW - Frontend selected TS modules (5e3cd6bf32) | Not emitted | 98.20 | 100.00 | 99.63 |
| HW - Staging tooling (release-core / sample / smoke) | Not emitted | 96.08 | 100.00 | 100.00 |
| HW - Load tooling (run.mjs / metrics.mjs) | Not emitted | 90.30 / 100.00 | 100.00 | 94.34 / 100.00 |
| MAD - Backend Jest/V8 (development 8b2c63f459) | 96.90 | 89.96 | 90.02 | 96.90 |
| MAD - Frontend Vitest/V8 (development 8b2c63f459) | 92.90 | 89.28 | 90.83 | 95.04 |
| Shared - API Gateway (develop f12aad6cb4) | Not emitted | 82.10 | 86.43 | 95.27 |
| Shared - WebSocket Service (development f12ef41ba9) | 91.42 | 86.88 | 85.36 | 96.19 |
| FDS - Backend Jest unit (develop 4b5ef57d4c) | 93.43 | 84.25 | 93.29 | 93.63 |
| FDS - Frontend Jest (develop 4b5ef57d4c) | 88.51 | 80.09 | 88.71 | 88.96 |
| FDS - QA snapshot (e590326 BE / 1065617 FE) | 96.19 / 89.96 | 87.15 / 81.33 | 96.85 / 92.41 | 96.23 / 89.85 |
| FOSS - Backend Jest unit (develop 0d80cf5c66) | 82.97 | 76.64 | 73.90 | 83.33 |
| FOSS - Backend QA snapshot (9ac1e6f live / CI) | 84.63 / 84.92 | 78.67 / 78.68 | 75.45 / 82.26 | 84.91 / 85.34 |
| FOSS - Critical Business Logic Core (28 suites) | 96.10 | 84.21 | 94.80 | 96.40 |
| FOSS - Frontend Node test runner (81 suites) | Not measured | Not measured | Not measured | Not measured |
| KMS - Backend Jest unit & E2E (develop 2bfa14b553) | Not measured | Not measured | Not measured | Not measured |
| KMS - Frontend Jest (develop 2bfa14b553) | Not measured | Not measured | Not measured | Not measured |
| WKMS - Backend Jest whole-app (develop 1a33059c00) | 88.72 | 85.45 | 82.44 | 91.25 |
| WKMS - Frontend Jest whole-app (develop 1a33059c00) | 88.53 | 81.14 | 91.50 | 89.74 |

### Test execution and traceability

| Subsystem | Reported passing automated tests | Selected register cases | Core qualifications |
|---|---|---|---|
| HW | 597 core + E2E/quality + 22 staging | 33 | 460 BE unit, 79 BE DB int (7 skipped), 58 FE Node, browser E2E & 48 quality audits pass |
| MAD (+ shared) | 1,087 MAD + 94 shared services | 58 | All MAD CI jobs & dependency audits pass; api-gateway (39), websocket-service (36), infra (19) pass |
| FDS | 949 (develop) / 903 (QA snapshot) | 43 | 531 BE unit, 56 DB int, 143 HTTP E2E, 219 FE Jest pass; 0 npm audit vulnerabilities |
| FOSS | 795 CI + 269 FE Node (557+269+164 QA) | 54 | 558 BE unit + 237 E2E (5 skipped) on 0d80cf5c66; 81 FE suites (269 passed, 1 skipped) on 9ac1e6f |
| KMS | 665 (6 skipped) | 82 | 234 BE unit (54 suites), 58 BE E2E (10 suites, 6 skipped), 373 FE Jest (59 suites); no --coverage in CI |
| WKMS | 649 (43 skipped) | 37 | 400 BE unit (55 suites, 9 skipped) & 249 FE Jest (33 suites) pass >=80% global coverage gates |

Selected test-register cases represent requirement-traced verification scenarios and explicit cross-service or manual acceptance boundaries rather than raw assertion totals. Separate coverage reruns (such as HW frontend coverage running 53 of 58 tests) are not double-counted in distinct test totals.

# 2. Hotel Website (HW)

Repository: forever-hotel/hotel-website | Branch: develop (merged PR #99 from pre-develop 440761c34a) | Tested revision: 5e3cd6bf323bc005e9efb86b5e09ba0e3015e727 (QA deliverable snapshot: 63673ba704b4ae645a8f067e246309b6fb0f03d0) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/hotel-website/actions/runs/42504096480, https://github.com/forever-hotel/hotel-website/actions/runs/42504096452, https://github.com/forever-hotel/hotel-website/actions/runs/42504621927, https://github.com/forever-hotel/hotel-website/actions/runs/42504621987, https://github.com/forever-hotel/hotel-website/actions/runs/42500427127

### Executed test groups

| Test group | Suites | Passed | Failed | Skipped |
|---|---|---|---|---|
| Backend Jest unit (develop 5e3cd6bf32) | 51 | 460 | 0 | 0 |
| Backend PostgreSQL integration (develop 5e3cd6bf32) | 17 passed; 1 skipped | 79 | 0 | 7 |
| Frontend Node unit tests (npm test) | 15 | 58 | 0 | 0 |
| Frontend coverage subset (npm run test:coverage) | Selected TS | 53 | 0 | 0 |
| Browser E2E journeys (npm run test:e2e) | 3 journeys | 3 | 0 | 0 |
| Browser responsive/accessibility quality (test:quality) | 5 viewports | 48 | 0 | 0 |
| Staging tooling (deploy/staging test:coverage) | 3 modules | 22 | 0 | 0 |
| Performance load tooling (performance test:coverage) | 2 modules | 8 | 0 | 0 |

### Measured coverage

| Measured scope | Statements | Branches | Functions | Lines |
|---|---|---|---|---|
| Backend Jest source scope (develop 5e3cd6bf32) | 88.38 | 82.10 | 87.14 | 88.39 |
| Backend Jest source scope (QA snapshot 63673ba704) | 88.02 | 82.16 | 87.50 | 88.05 |
| Frontend selected TS modules (develop 5e3cd6bf32) | Not emitted | 98.20 | 100.00 | 99.63 |
| Frontend selected TS modules (QA snapshot 63673ba704) | Not emitted | 98.49 | 100.00 | 99.75 |
| Staging tooling (release-core / sample-core / smoke) | Not emitted | 96.08 | 100.00 | 100.00 |
| Load tooling (performance/run.mjs & metrics.mjs) | Not emitted | 90.30 / 100.00 | 100.00 | 94.34 / 100.00 |

### Scope and interpretation

- On develop (5e3cd6bf32 / 440761c34a, 9 November 2026), backend unit coverage meets all four 80% global Jest thresholds (88.38% statements, 82.10% branches, 87.14% functions, 88.39% lines across 51 suites and 460 tests), while disposable PostgreSQL integration passes 79 tests across 17 suites (with 1 suite / 7 tests skipped when gated).
- Critical module branch assessment in the HW QA package (63673ba704) shows quote-pricing.ts (95.83%), payhere-payments.service.ts (91.48%), and payhere-provider.ts (96.22%) meeting the 90% critical-branch target, whereas bookings.service.ts (87.50%) and booking-management.service.ts (84.54%) remain slightly below 90%, and SQL repositories are primarily exercised by PostgreSQL integration suites rather than unit mocks.
- Frontend coverage instruments imported TypeScript logic modules (53 tests in test:coverage) and excludes TSX rendering; browser UI rendering and responsive/keyboard behavior are verified separately by npm run test:e2e (3 full browser journeys) and npm run test:quality (48 viewport/accessibility checks across 320px, 375px, 768px, 1024px, and 1440px).

### CI and reproduction evidence

https://github.com/forever-hotel/hotel-website/actions/runs/42504096480 |
https://github.com/forever-hotel/hotel-website/actions/runs/42504096452 |
https://github.com/forever-hotel/hotel-website/actions/runs/42500427127 | Reproduction: backend: npm run test:cov -- --runInBand; npm run test:integration; frontend: npm test; npm run test:coverage; npm run test:e2e; npm run test:quality.

# 3. Manager Dashboard (MAD)

Repository: forever-hotel/manager-dashboard (+ forever-hotel/api-gateway, forever-hotel/websocket-service, forever-hotel/infra) | Branch: development / develop | Tested revision: 8b2c63f459a93c217eb7cb502a2500e80181aa6e (api-gateway: f12aad6cb4; websocket-service: f12ef41ba9; infra: e3e17fcaac) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/manager-dashboard/actions/runs/42480400290,
https://github.com/forever-hotel/manager-dashboard/actions/runs/42504148805,
https://github.com/forever-hotel/api-gateway/actions/runs/42504067779,
https://github.com/forever-hotel/websocket-service/actions/runs/42504249365,
https://github.com/forever-hotel/infra/actions/runs/42501357369

### Executed test groups

| Test group | Suites | Passed | Failed | Skipped |
|---|---|---|---|---|
| MAD Backend Jest unit (npm run test:cov) | 42 | 534 | 0 | 0 |
| MAD Backend changed-code coverage check | 1 | 5 | 0 | 0 |
| MAD Backend PostgreSQL & RabbitMQ integration | 15 | 203 | 0 | 0 |
| MAD Central auth & shared WebSocket system check | 1 | 1 | 0 | 0 |
| MAD Frontend Vitest unit/component (test:cov) | 38 | 341 | 0 | 0 |
| MAD Frontend build & audit prechecks | 2 | 6 | 0 | 0 |
| MAD Frontend Playwright E2E (npm run test:e2e) | 1 | 8 | 0 | 0 |
| Shared API Gateway (forever-hotel/api-gateway) | Node test | 39 | 0 | 0 |
| Shared WebSocket Service (forever-hotel/websocket-service) | 6 | 36 | 0 | 0 |
| Shared Infrastructure Compose/release checks (infra) | Node test | 19 | 0 | 0 |

### Measured coverage

| Measured scope | Statements | Branches | Functions | Lines |
|---|---|---|---|---|
| MAD Backend Jest/V8 (8b2c63f459) | 96.90 | 89.96 | 90.02 | 96.90 |
| MAD Frontend Vitest/V8 (8b2c63f459) | 92.90 | 89.28 | 90.83 | 95.04 |
| MAD Historical baseline (bba65a0762, 24 Oct 2026) | 97.08 / 94.27 | 90.69 / 91.25 | 92.24 / 93.34 | 97.08 / 96.02 |
| Shared API Gateway (f12aad6cb4, npm run test:cov) | Not emitted | 82.10 | 86.43 | 95.27 |
| Shared WebSocket Service (f12ef41ba9, npm run test:cov) | 91.42 | 86.88 | 85.36 | 96.19 |

### Scope and interpretation

- The previous backend dependency audit failure noted in the 25 October 2026 report was resolved by pinning transitive overrides (multer, body-parser, qs) in backend/package.json; all jobs in Runs 42480400290 and 42504148805 passed with zero high/critical vulnerabilities.
- Global line and branch thresholds of 80% and critical-module / changed-code checks are enforced in CI. MAD executes 1,087 core tests (534 backend unit, 203 backend integration, 1 system integration, 341 frontend Vitest, 8 Playwright E2E) plus 11 Node verification tests.
- Shared platform services originally prototyped alongside MAD are now maintained and verified in dedicated organization repositories: forever-hotel/api-gateway (39 passing tests, 95.27% lines / 82.10% branches), forever-hotel/websocket-service (36 passing tests across 6 suites, 96.19% lines / 86.88% branches), and forever-hotel/infra (19 passing Compose and release checks).

### CI and reproduction evidence

https://github.com/forever-hotel/manager-dashboard/actions/runs/42480400290 | https://github.com/forever-hotel/api-gateway/actions/runs/42504067779 | https://github.com/forever-hotel/websocket-service/actions/runs/42504249365 | Reproduction: backend: npm run test:cov -- --runInBand; npm run test:integration; npm run test:system; frontend: npm run test:cov; npm run test:e2e.

# 4. Front Desk System (FDS)

Repository: forever-hotel/front-desk | Branch: develop (QA evidence branch: feature/84-fds-final-qa-evidence) | Tested revision: 4b5ef57d4c4ce03c9ab2fe78f42b94eff96838cd (QA snapshot: e590326 backend / 1065617 frontend / a6050dd docs) | CI date: 9 November 2026 (QA snapshot: 27 October 2026)

Evidence: https://github.com/forever-hotel/front-desk/actions/runs/42506217082, https://github.com/forever-hotel/front-desk/actions/runs/42506217056, https://github.com/forever-hotel/front-desk/actions/runs/42506217053 (QA snapshot runs: 40530366600, 40534533200, 40534533251)

### Executed test groups

| Test group | Suites | Passed | Failed | Skipped |
|---|---|---|---|---|
| Backend Jest unit (develop 4b5ef57d4c) | 72 | 531 | 0 | 0 |
| Backend PostgreSQL integration (develop 4b5ef57d4c) | 10 | 56 | 0 | 0 |
| Backend HTTP/Socket.IO E2E (develop 4b5ef57d4c) | 12 | 143 | 0 | 0 |
| Frontend Jest (develop 4b5ef57d4c) | 50 | 219 | 0 | 0 |
| QA snapshot baseline (e590326 BE / 1065617 FE) | 129 total | 903 | 0 | 0 |

### Measured coverage

| Measured scope | Statements | Branches | Functions | Lines |
|---|---|---|---|---|
| Backend Jest source scope (develop 4b5ef57d4c) | 93.43 | 84.25 | 93.29 | 93.63 |
| Frontend Jest source scope (develop 4b5ef57d4c) | 88.51 | 80.09 | 88.71 | 88.96 |
| Backend Jest source scope (QA snapshot e590326) | 96.19 | 87.15 | 96.85 | 96.23 |
| Frontend Jest source scope (QA snapshot 1065617) | 89.96 | 81.33 | 92.41 | 89.85 |

### Scope and interpretation

- On develop head 4b5ef57d4c (9 November 2026, PR #101), all 949 automated CI tests pass across backend unit (531), PostgreSQL integration (56), HTTP/Socket.IO E2E (143), and frontend Jest (219), and all four coverage metrics in both backend and frontend exceed the configured 80% Jest threshold.
- Both backend and frontend npm audit checks report 0 vulnerabilities following the upgrade of undici to 8.12.2.
- In addition to CI suites, the committed FDS QA evidence package (feature/84-fds-final-qa-evidence @ a6050dd) records bounded local verification runs: Core lifecycle Plan 02 (35 PASS / 0 FAIL), Realtime baseline Plan 03 (14 PASS / 5 BLOCKED), Phase B real WKMS boundary (3 PASS / 3 BLOCKED), Phase C broker restart (5 PASS / 3 BLOCKED), Phase D browser UX (8 PASS / 0 FAIL), W04-B manual WKMS assignment (6 PASS / 2 BLOCKED), and read-only API performance (15 PASS / 1,800 requests).

### CI and reproduction evidence

https://github.com/forever-hotel/front-desk/actions/runs/42506217082 |
https://github.com/forever-hotel/front-desk/actions/runs/42506217056 |
https://github.com/forever-hotel/front-desk/actions/runs/40530366600 | Reproduction: backend: npm run test:cov -- --runInBand; npm run test:integration; npm run test:e2e; frontend: npm run test:cov.

# 5. Food Ordering and Service Requests (FOSS)

Repository: forever-hotel/food-ordering | Branch: develop (QA evidence package committed on develop from docs/111-foss-qa-evidence) | Tested revision: 0d80cf5c66b566580927cb8fae1f7d549a15f03f (QA snapshot: 9ac1e6f3df9f93ee76be6db479dc8c4bf6529328) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/food-ordering/actions/runs/42490527447, https://github.com/forever-hotel/food-ordering/actions/runs/41833172660

### Executed test groups

| Test group | Suites | Passed | Failed | Skipped |
|---|---|---|---|---|
| Backend Jest unit & integration (develop 0d80cf5c66) | 63 | 558 | 0 | 0 |
| Backend E2E suites on CI PostgreSQL (develop 0d80cf5c66) | 19 passed; 1 skipped | 237 | 0 | 5 |
| Backend Jest unit & integration (QA snapshot 9ac1e6f) | 63 | 557 | 0 | 0 |
| Critical Business Logic Core subset (QA snapshot 9ac1e6f) | 28 | 312 | 0 | 0 |
| Frontend Node native test runner (npm --prefix frontend test) | 81 | 269 | 0 | 1 |

### Measured coverage

| Measured scope | Statements | Branches | Functions | Lines |
|---|---|---|---|---|
| Backend Jest global scope (develop 0d80cf5c66, Run 42490527447) | 82.97 | 76.64 | 73.90 | 83.33 |
| Backend Jest live branch run (QA snapshot 9ac1e6f) | 84.63 | 78.67 | 75.45 | 84.91 |
| Backend Jest CI baseline (9ac1e6f, Run 41833172660) | 84.92 | 78.68 | 82.26 | 85.34 |
| Critical Business Logic Core (28 suites, 312 tests) | 96.10 | 84.21 | 94.80 | 96.40 |
| Frontend Node test runner (81 suites, 269 passed) | Not measured | Not measured | Not measured | Not measured |

### Scope and interpretation

- Between 13 October 2026 (18 develop tests, 60.37% line coverage) and 9 November 2026 (develop @ 0d80cf5c66 / 9ac1e6f), FOSS expanded to 63 backend unit suites (558 passed), 20 backend E2E suites (237 passed, 5 skipped on disposable CI PostgreSQL), and 81 frontend Node test suites (269 passed, 1 skipped).
- Configured global gates in backend/jest.config.ts (80% statements, 80% lines, 75% functions, 70% branches) were all met on the 9ac1e6f QA snapshot (84.63% statements, 78.67% branches, 75.45% functions, 84.91% lines); on commit 0d80cf5c66 (PR #113 menu fallback), statements (82.97%), lines (83.33%), and branches (76.64%) exceed 80%/70% while global functions is 73.90%. Core business modules (guest-sessions 96.20% lines, food-orders 93.80%, service-requests 95.20%, complaints 93.10%, menu 91.50%, auth 96.50%) achieve 96.40% line and 84.21% branch coverage (5.79% below the 90% stretch branch target).
- Frontend tests execute via Node.js v24 native test runner (node --test --experimental-strip-types) without global V8 percentage emission, and 5 Playwright browser specs in frontend/tests/e2e/ remain unverified in CI.

### CI and reproduction evidence

https://github.com/forever-hotel/food-ordering/actions/runs/42490527447 | https://github.com/forever-hotel/food-ordering/actions/runs/41833172660 | Reproduction: backend: npm --prefix backend run test:cov -- --runInBand; npm --prefix backend run test:e2e; frontend: npm --prefix frontend test.

## 6. Kitchen Management System (KMS)

Repository: forever-hotel/kitchen-management | Branch: develop | Tested revision: 2bfa14b5535c5c065f49dbdc2b7c62c4c8105d15 (PR #101) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/kitchen-management/actions/runs/42502180570 (Historical Oct 10 baseline: 89087002ac2df8f07b6140c00111979c75bc2799, Run 38038213541)

### Executed test groups

| Test group | Suites | Passed | Failed | Skipped |
|---|---|---|---|---|
| Backend Jest unit (npm test, develop 2bfa14b553) | 54 | 234 | 0 | 0 |
| Backend Jest E2E (npm run test:e2e, develop 2bfa14b553) | 10 passed; 2 skipped | 58 | 0 | 6 |
| Frontend Jest (npm run test:ci, develop 2bfa14b553) | 59 | 373 | 0 | 0 |
| Historical develop baseline (89087002ac, 10 Oct 2026) | 28 total | 76 | 0 | 0 |

### Measured coverage

| Measured scope | Statements | Branches | Functions | Lines |
|---|---|---|---|---|
| Backend unit/E2E (develop 2bfa14b553) | Not measured | Not measured | Not measured | Not measured |
| Frontend Jest (develop 2bfa14b553) | Not measured | Not measured | Not measured | Not measured |

### Scope and interpretation

- Automated test execution in forever-hotel/kitchen-management expanded from 76 passing tests across 28 suites (10 October 2026) to 665 passing tests (and 6 skipped Postgres-specific E2E tests) across 123 passing suites on 9 November 2026 (Run 42502180570 on commit 2bfa14b553), covering kitchen order creation, reading, state transitions, allergy safety, ordering windows, and menu item/category CRUD.
- Both backend and frontend security dependency audits (npm run security:audit) passed with 0 vulnerabilities in Run 42502180570.
- Although test:cov scripts are defined in backend/package.json and frontend/package.json, the GitHub Actions workflows (.github/workflows/backend-ci.yml and frontend-ci.yml) invoke npm test, npm run test:e2e, and npm run test:ci without --coverage or a configured coverageThreshold gate. Coverage percentages remain unmeasured in CI.

### CI and reproduction evidence

https://github.com/forever-hotel/kitchen-management/actions/runs/42502180570 | Reproduction / configured coverage command: backend/frontend: npm run test:cov -- --runInBand (not invoked in the cited CI workflow).

# 7. Worker Management System (WKMS)

Repository: forever-hotel/worker-management | Branch: develop | Tested revision: 1a33059c0005752ca709ca91ab11067dc6cff55a (PR #75 82f458005b; Oct 25 baseline: 0f91f1c9d6390c80c9f12f82b834fee9bf374104) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/worker-management/actions/runs/42504698971, https://github.com/forever-hotel/worker-management/actions/runs/42504970394, https://github.com/forever-hotel/worker-management/actions/runs/42496517266 (Oct 25 baseline: 40174879698)

### Executed test groups

| Test group | Suites | Passed | Failed | Skipped |
|---|---|---|---|---|
| Backend Jest with coverage (develop 1a33059c00) | 55 passed; 9 skipped | 400 | 0 | 43 |
| Frontend Jest with coverage (develop 1a33059c00) | 33 | 249 | 0 | 0 |
| Historical Oct 25 baseline (0f91f1c9d6) | 56 passed; 7 skipped | 445 | 0 | 33 |

### Measured coverage

| Measured scope | Statements | Branches | Functions | Lines |
|---|---|---|---|---|
| Backend Jest whole-application (develop 1a33059c00) | 88.72 | 85.45 | 82.44 | 91.25 |
| Frontend Jest whole-application (develop 1a33059c00) | 88.53 | 81.14 | 91.50 | 89.74 |
| Backend Jest source scope (Oct 25 baseline 0f91f1c9d6) | 93.28 | 88.19 | 89.30 | 94.80 |
| Frontend Jest source scope (Oct 25 baseline 0f91f1c9d6) | 93.81 | 85.55 | 94.21 | 94.90 |
| Realtime gateway targeted run (db7d682c96 src/realtime/**) | 91.93 | 90.00 | 90.00 | 93.33 |

### Scope and interpretation

- On develop head 1a33059c00 (9 November 2026), WKMS expanded from 445 to 649 passing automated tests (400 backend + 249 frontend) while enforcing >=80% global branch and line coverage gates in both backend (85.45% branches, 91.25% lines) and frontend (81.14% branches, 89.74% lines).
- Nine backend suites (43 tests) requiring external live infrastructure remain skipped in unit/coverage CI runs and are reported explicitly rather than counted as passes.
- Frontend, Backend, and Docker image build jobs all succeeded with 0 dependency vulnerabilities; only the optional post-merge immutable registry publishing job on develop push failed env validation because public release URLs were not injected into non-release CI.

### CI and reproduction evidence

https://github.com/forever-hotel/worker-management/actions/runs/42504698971 | https://github.com/forever-hotel/worker-management/actions/runs/42504970394 | Reproduction: backend: npm run test:cov -- --runInBand; frontend: npm run test:coverage.

## 8. Evidence interpretation and remaining actions

- Enable --coverage and enforce 80% coverage thresholds in forever-hotel/kitchen-management CI (.github/workflows/backend-ci.yml and frontend-ci.yml) so KMS reports instrumented coverage alongside its 665 passing tests.
- Add V8/c8 coverage reporting to the forever-hotel/food-ordering frontend Node test runner (81 suites, 269 passing tests) and raise FOSS critical-logic branch coverage from 84.21% to the 90% stretch target and HW booking service branch coverage from 84.54%–87.50% to 90%.
- Retain dated LCOV/HTML coverage artifacts and CI logs for the exact tagged release candidate across all six subsystem repositories and the three shared service repositories (api-gateway, websocket-service, infra).
- Distinguish unit/component coverage and disposable-database CI integration from live cross-subsystem staging acceptance, external payment/email/printer hardware verification, and hotel-wide load testing.

Passing unit, component, or mock-backed integration suites establish subsystem code quality within their instrumented boundaries and must not be represented as signed-off multi-service UAT or production release certification.